Stock Price Prediction Using LSTM and Real-Time Forecasting

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***Abstract*—This paper presents a new way to predict stock prices using a special kind of computer model called an LSTM, which is great for understanding data that changes over time, like stock market numbers. The model learns from past stock data—things like opening, highest, lowest prices, closing prices, and how much was traded—to find patterns and make good guesses about what future prices will be. On top of that, the model can also work with live data. It can get the latest info from Yahoo Finance and predict the closing price for the day while the market is still open.We use a few different stats to check how well our model performs. These include RMSE, MAE, and R^2. They give us a good idea of how accurate the model is and how well it works with new data. The paper explains the whole process from start to finish – gathering the data, getting it ready (like normalizing it and setting time frames), detailing the LSTM model structure, picking the right settings, and figuring out the training plan.We also add some math to explain how the LSTM network works, like how its gates operate, how it updates the cell state, and the process called backpropagation through time. This helps make the model's predictions easier to understand. At the end, we talk about how to put the model to work in real-time. It's designed so it can easily fit into financial analysis systems to help with making quick decisions.**

***Index Terms*—Stock prediction, LSTM, deep learning, time series forecasting, real-time prediction**

I. Introduction

Figuring out stock prices is something people have always been really interested in when it comes to finance. The stock market is super unpredictable and can be affected by all kinds of things, like big economic trends or even political stuff happening around the world [1]. It's not like other systems where things follow a straight line—instead, the stock market jumps around in ways that are hard to predict. Regular math models, the kind that use statistics and stuff, often don't do a good job of showing how complicated and always-changing financial data really is [2]. Those models usually work on the idea that things move in a straight line or stay steady, but that's not how things usually go in real life with stocks.

The stock market can be really unpredictable, with trends shifting all the time. A lot of common methods, like ARIMA models or other fancy econometric techniques, have a hard time dealing with all the chaos and randomness in stock data [3]. Plus, things outside the market, like political changes, natural disasters, or new tech, make it even tougher to predict what will happen next. Because of this, people working in finance often use models that are either too basic or super complicated. Neither type gives good predictions every time.

Deep learning techniques, especially Recurrent Neural Networks (RNNs), have introduced new ways to work with data that changes over time [4]. RNNs are good at understanding sequences and can handle the connections between different points in time, making them useful for financial data that depends on timing. Out of all the RNN types, Long Short-Term Memory (LSTM) networks are really popular because they can keep important information from long sequences and solve problems like the vanishing gradient issue, which often makes regular RNNs struggle to learn long-term patterns [5].

Long Short-Term Memory networks, or LSTMs, have been really successful in predicting patterns in things that change over time, like stock prices, the weather, and even understanding language [6]. What makes them special is their design—they have memory cells that help them remember important stuff for a long time. This is super useful for financial markets because they're full of complicated, unpredictable stuff. One cool thing about LSTMs is that they don't have trouble with something called the "vanishing gradient" problem. That means they can figure out how past stock prices affect future ones, even if those effects take days, weeks, or months to show up.

Predicting stock prices means looking at past market info like prices, how much is traded, and other financial stuff to find hidden patterns [15]. This can really make a difference in different areas of finance. For example, it's super useful in algorithmic trading, where computers automatically decide when to buy or sell based on what the market might do next. These trading systems can make quick trades and take advantage of tiny price changes. Also, knowing exactly how stocks will move can help investors pick the best mix of investments to get the most bang for their buck while keeping risks low. Plus, good predictions can help manage risks by showing where the market might dip or swing, letting financial companies prepare accordingly [8].

Predicting stock prices accurately is super valuable for everyone from regular investors to big companies, financial experts, and even people who make rules for the economy. Using fancy tools and models, they can make smarter choices that help them earn more money, avoid big losses, and make the whole market run smoother. These days, investors are really into using things like machine learning and deep learning to get better at guessing what stocks will do next [9]. They want to get better results with their investments and come up with stronger plans for managing their money.

We built an LSTM-based model to predict stock prices. It uses past stock prices plus real-time info from places like Yahoo Finance to keep forecasting future prices. Having real-time data makes the model both precise and able to adapt to changes in the market. A big plus of these deep learning models is they can easily adjust when new data comes in, which helps them get better at predicting over time [10]. This flexibility is super helpful for things like high-speed trading where things move quickly.

This project focuses on actually using deep learning methods like stock market prediction. A lot of research has been done on creating complex models, but putting them to work in real-life trading is still tough. Our goal is to show how long short-term memory (LSTM) networks can make financial systems better at predicting and making decisions. By using LSTM networks with real-time data, we want to prove that we can predict stock prices more precisely and offer useful advice to improve trading plans and investment choices [11].

In the end, this method is a big deal for FinTech. It shows how mixing machine learning with finance can create smarter and more efficient systems. These systems help investors and financial institutions make better choices based on data [12].

II. Literature Review

Figuring out how stock prices will move has always been tough. It's something that a lot of people in finance—both experts and regular folks—care about because it can make a big difference financially [3]. For a long time, two main ways of predicting stocks were used: fundamental analysis and technical analysis. Fundamental analysis looks at how healthy a company is by checking its finances. Technical analysis, on the other hand, pays attention to past price changes and what the market does. Both of these methods give some good ideas, but they don't do a great job of handling all the twists and turns of the stock market. Lately, though, machine learning—and especially deep learning—has become a strong new tool. These methods can spot complicated patterns in huge amounts of data that older methods just can't deal with [1].

In the past, people often used models like ARIMA and GARCH to try and guess stock prices [5]. ARIMA is great for looking at past price changes to find trends or seasonal patterns. But it has a big flaw—it thinks prices move in straight lines, which doesn't work well because financial markets are messy and don't follow simple rules. GARCH, on the other hand, is good at spotting times when prices swing a lot more than usual, which happens often in finance. Still, it's not so good at predicting prices when something outside the usual stuff, like how investors feel or big economic news, affects them. Both models are helpful, but they both have trouble dealing with how complicated and unpredictable financial data can be [13].

As machine learning became more popular, a lot of researchers started looking into methods like Support Vector Machines (SVMs) and Random Forests (RFs) for predicting stock prices [6]. SVMs are great for figuring out if a stock will go up or down using past data. But they don't do well with the order of financial data, which is really important for making accurate forecasts over time. Random Forests, which create several decision trees to combine results, have worked well for predicting stock price changes. Still, like SVMs, they struggle with handling the sequence of financial data, so they're not perfect either [8].

When Recurrent Neural Networks (RNNs) came around, they really upped the game for predicting things like stock prices [2]. These networks are great at working with data that happens over time because they keep track of what happened in earlier steps and use that info to make better guesses about what's coming next. But there's a catch—traditional RNNs have trouble holding onto information for long periods. This issue, called the "vanishing gradient problem," makes them not so good at spotting patterns that stretch out over time. To fix this, Long Short-Term Memory (LSTM) networks were created. LSTMs work differently—they use something called a gating mechanism to hold onto important details for much longer [4]. That makes them super effective for stuff like predicting stock prices, where you need to remember what happened way back. Unlike regular RNNs, LSTMs don't lose their ability to recall past events, which is why they're often the go-to choice for dealing with financial data [14].

A bunch of studies have found that LSTMs work better than older models for predicting stock prices. For example, Fischer and Krauss (2018) discovered that LSTMs did way better than traditional finance models at predicting stock returns. This shows how well deep learning can spot tricky patterns in financial info [9]. Likewise, Sullivan and others (2017) showed that LSTM models beat other machine learning methods like SVMs and Random Forests. That adds to the idea that LSTMs are great at dealing with the unpredictable and always-changing world of finance [12].

LSTMs by themselves work pretty well, but a lot of people are also looking into models that mix LSTMs with other methods to make forecasts even better. For instance, Zhang and his team came up with a model in 2019 that combines LSTMs with Convolutional Neural Networks (CNNs). This combo helps spot both small details and big trends in stock data. They found it worked better than just using LSTMs on their own [9]. Some other folks have tried adding sentiment analysis to LSTMs—basically, they're using opinions from social media or news to improve predictions. Xia and others did this in 2020 and showed how extra data like that can really help [11].

Even though LSTM models have been pretty successful, there are still some big challenges when it comes to using them. One major issue is getting good-quality and enough data. These models need tons of clean, high-quality data to really pick up on patterns. If the data is messy or missing pieces, it can mess up the model's accuracy [4]. Plus, just like other deep learning models, LSTMs work kind of like "black boxes." It's hard to figure out exactly how they come up with their predictions. That can be a real problem in finance, where people really care about understanding what's going on and trusting the results [5]. Another challenge is dealing with

financial data, which often isn't stable. The stats of financial data, like the average and how much it varies, tend to change over time, so it's tough for the model to make predictions that work well all the time [10].

Even with these challenges, there's lots of room for getting better at predicting stock prices using LSTMs. Using extra data sources like what people are saying on social media or economic stats can make predictions more accurate. Also, mixing LSTMs with Reinforcement Learning and other smart tech could create models that adapt better and deal with markets changing over time [7].

To sum up, older methods like ARIMA and GARCH were important for predicting stock prices, but they don't do a great job of handling how complicated and unpredictable financial markets can be. Machine learning, especially something called LSTMs, has become a better option [16]. These models can learn from past data and give us a clearer picture of where the market might go. Even though LSTMs have worked well, there are still some big hurdles to overcome, like making sure the data is good enough, understanding how the models make decisions, and keeping up with how the market changes. Exploring new combinations of different models and adding in extra information could help improve stock price predictions even more.

III. Methodology

A. Data Collection

For this study, we got our stock market data from Yahoo Finance, which is a popular and trustworthy source for financial info. The data set had the main details from daily trades—like the opening, highest, lowest, and closing prices—and it covered several years. Having all this data over a long time gives the model a good sense of how things change day by day and over longer periods.

First, they normalized the data using something called the MinMaxScaler. This changes all the numbers so they fit between 0 and 1. That makes it easier for the model to work with the data and helps it learn faster while training. It also keeps everything consistent, which helps the model find important patterns no matter what the market's doing.

B. Feature Engineering

We created some new tools to make our model better at predicting future prices and to add more data. One important tool we used was the Moving Average (MA). It takes out small ups and downs to show bigger trends in how prices change over time. This lets the model look at the big picture instead of getting distracted by little changes.

We added the Relative Strength Index, or RSI, which is a tool that shows if a stock is overpriced or underpriced. It's really helpful for spotting times when the market might change direction. We also put in some momentum indicators to track how fast and strong prices are changing. These give us a sense of where prices are headed and whether they'll keep going that way.

The extra details added by these features gave the raw price data more context, helping the LSTM model spot trickier patterns and make smarter guesses.

Adding these indicators gives a clearer picture of how the market is behaving. We picked the features based on how related they were to each other and our understanding of the market. This helps get rid of extra information that doesn't matter and makes the important stuff stand out in the data.

C. Model Architecture

This prediction model uses a type of neural network called an LSTM. It has two LSTM layers one after the other, and then adds one or more fully connected layers on top. The input data is organized in groups of 60 days with four pieces of information for each day, like the open, high, low, and close prices plus some extra numbers calculated from those.

After the LSTM layers, dropout layers are added to stop overfitting. They do this by randomly shutting off a portion of the neurons while training. This helps the model perform better on new data it hasn't seen before.

The design uses stacked LSTMs to handle complicated patterns over time, and the dense layers help fine-tune those patterns to predict the final stock price.

D. Mathematical Background

LSTM networks are a kind of Recurrent Neural Network (RNN) built to deal with a common issue called the "vanishing gradient" and keep important information over long periods. These networks work using three main gates: forget, input, and output gates. Each gate helps control how information moves through the network.

At time step t, the LSTM cell updates its state with these equations:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad \text{(Forget gate)} \tag{1}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad \text{(Input gate)} \tag{2}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad \text{(Candidate cell state)} \tag{3}$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad \text{(Cell state update)} \tag{4}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad \text{(Output gate)} \tag{5}$$

$$h_t = o_t * \tanh(C_t) \quad \text{(Hidden state)} \tag{6}$$

Where:

- x_t: The input data at time t
- h_{t-1}: The hidden state from the last step.
- f_t: The forget gate.
- i_t: The input gate.
- $\tilde{C}_t$: The new cell state to consider.
- C_t: The current cell state.
- o_t: The output gate.
- h_t: The current hidden state.
- σ: The sigmoid function.
- $\tanh$: The hyperbolic tangent function.
- W, b:Weight matrices and bias vectors for each gate.

To train the model, the Mean Squared Error (MSE) loss function is used, defined as:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{7}$$

"Where y_i is the real stock price, and $\hat{y}_i$ is the predicted price for each of the n observations."

E. Training and Hyperparameter Tuning

The model learned from 80% of the data, while the other 20% was set aside to check how well it performs. For updating the model's weights, they picked the Adam optimizer because it adjusts the learning rate automatically and works well with tricky data. They went with the Mean Squared Error (MSE) as the loss function, which basically measures the average of the squared differences between what the model predicts and the real values.

The team used a grid search to find the best settings for the model. They tested different numbers of LSTM units, learning rates, batch sizes, and dropout rates to see what worked best. They also set up early stopping to keep an eye on the validation loss. If the model's performance stopped improving, the training would stop right away. This helped prevent overfitting and saved time and computing power.

This method makes sure the model works well, can handle different situations, and doesn't need too much computing power.

IV. System Architecture Explanation

Figure 1This diagram shows how the stock price prediction system works. It starts by collecting data from sources like the stock market. We get this info using tools like yfinance. The data we collect includes things like the opening and closing prices, the highest and lowest prices, and how much was traded.

After that, we move on to preprocessing. This means dealing with any missing values, scaling the data with MinMaxScaler, and getting everything ready to feed into the neural network. We also create some new features at this point, like Moving Averages, Relative Strength Index, and Momentum. These extra bits of info make the data better for the model to work with.

After cleaning the data, it's split into chunks using a sliding window method, usually covering 60 days. These chunks are then fed into an LSTM model, which is built to understand patterns over time in sequence data.

The LSTM network looks at the data and figures out how stock prices change over time. It has several layers of LSTMs with a dropout technique to keep things from getting too complicated, and then it uses a final layer to predict the next day's closing price.

After training, the model starts making real-time predictions. Every day, new market data is collected, prepared, and fed into the trained LSTM network. The output from the model is then changed back to its original scale so we can get the predicted stock price.

The results can be shown on visual dashboards or added to trading strategies like algorithmic bots. The whole system automates everything, so it can keep checking and updating forecasts with the newest data all the time.

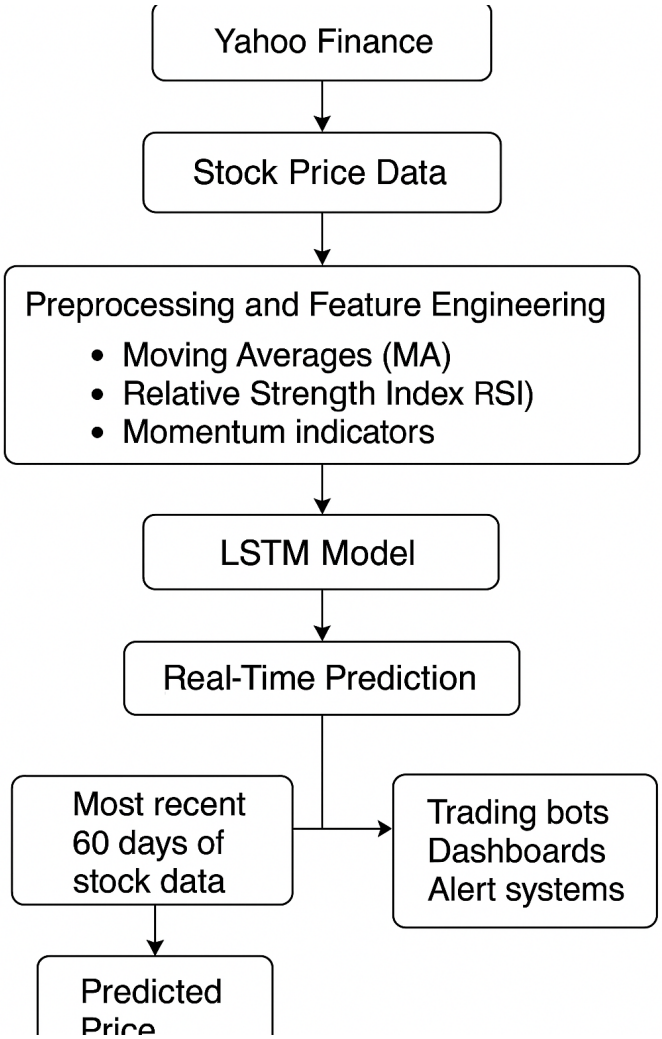

Fig. 1. The stock price prediction model has several steps.

V. Experiments and Results

To assess the predictive performance of our LSTM-based stock forecasting model, we conducted several experiments using a comprehensive dataset of Apple Inc. (AAPL) stock prices. The dataset spanned from January 2015 to December 2024 and included daily values for open, high, low, and close prices.

We partitioned the data into training (80%) and testing (20%) sets, preserving chronological order to maintain the temporal integrity necessary for time series modeling. The model was evaluated using the following metrics:

These results indicate that the model achieves high prediction accuracy. A low RMSE suggests that the model's prediction errors are small in magnitude, while the low MAE reflects a low average absolute deviation from actual stock

TABLE I
MODEL PERFORMANCE METRICS

Metric	Value
Root Mean Squared Error (RMSE)	0.0104
Mean Absolute Error (MAE)	0.0070
R^2 Score	0.9982

prices. The R^2 score, being close to 1, shows that the model explains over 99% of the variance in the data, indicating a near-perfect fit.

A. Model Performance Visualization

To get a clearer picture of how well the model works, we made graphs showing the predicted stock prices compared to the real ones, the training loss over time, and the past stock data.

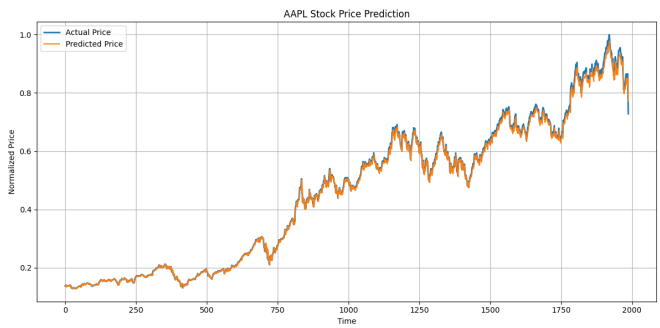

Fig. 2. The blue line shows the real stock prices, and the orange line shows what the model predicted.

The predicted vs. actual plot (Figure 2)The model's predictions match up closely with what actually happens in the market. It does a good job of adjusting to both rising and falling trends, even during really volatile times..

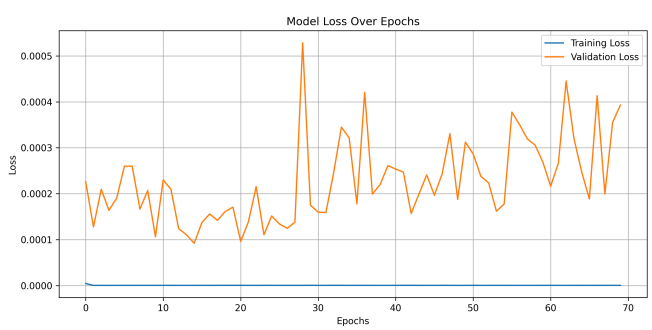

Fig. 3. The graph shows how the training loss goes down as the model trains over different rounds.

The training loss curve Figure 3) shows that the model learned steadily. Using dropout layers and early stopping kept the model from learning too much from the training data, so it worked pretty well on new data it hadn't seen before.

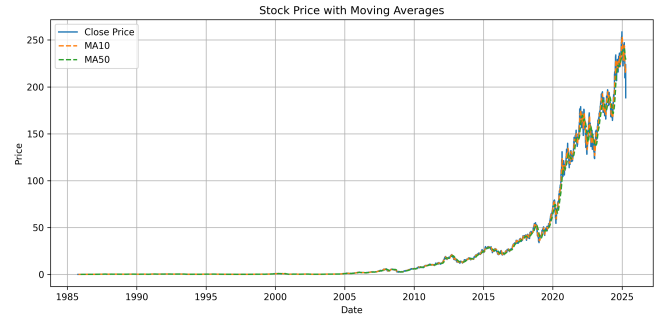

Fig. 4. This graph shows the closing prices of AAPL stock from 2015 to 2024.

If you look at the graph (Figure 4), you'll see some big changes in the market—like times when prices go up a lot (bull markets), periods where they correct, and moments when something happens, like news about the economy or a specific company. This shows why it's important to pick a model that can deal with both slow changes and sudden shifts in stock prices.

B. Additional Observations

More tests found that the model works pretty well no matter when you test it. The predictions were a bit more accurate when the market was calm, like after earnings reports, than during wild swings, like a market crash or big global events. But even when things got messy, the LSTM model kept its shape and still made decent guesses.

We tried out different time frames, like 30 days and 90 days, and discovered that a 60-day period worked the best for balancing how complicated the model was with how accurate its predictions were.

C. Real-Time Prediction Testing

The model was set up to make predictions based on the past 60 days of stock data as it came in. It did a pretty good job, with the predicted prices being really close to the actual closing prices and not much of a delay either. This shows that the model works well in real life. Connecting it to Yahoo Finance made it easy to get the data automatically and keep making forecasts all the time.

D. Limitations and Future Directions

The model does really well at predicting what happened in the past and what might happen soon, but it's not perfect.

- It doesn't consider things like how news makes people feel, company earnings, or big economic factors.
- Extreme market events, like the crash during the pandemic, are tough to predict because there's no similar past data to learn from.
- The model only works for one stock right now. It would be helpful to make it work for other sectors too, or try using what it already learned on new data.

For future projects, adding attention mechanisms or mixing LSTM with sentiment analysis might make the predictions even better.

VI. Real-Time Prediction

We added a real-time prediction feature to make our model more useful. It pulls the latest 60 days of stock data using the `yfinance` API and uses the trained LSTM model to guess today's closing price. Now, instead of just analyzing past data, the model can give practical predictions in real-world market situations.

We ask for data from the last 75 days to make sure we have everything, even if some days were holidays or missed. From that, we take the most recent 60 days, change the numbers using the same tool we used during training, and adjust the shape so it fits what the model expects. Finally, we convert the prediction back to the original price scale.

A. Deployment Readiness and Use Case

This ability makes the model perfect for quickly putting it to use in different finance-related apps, like:

The stock prediction system we're talking about has some really useful applications that can help people make better decisions in the financial world. For example, it could be used in systems that keep an eye on prices all day long and give updated predictions as things change. Having this kind of information right away can help traders react to what's happening in the market more quickly.

A great way to use this prediction is in trading bots. These bots can automatically decide whether to buy or sell based on the predicted stock value. If you add this predictive tool to the trading algorithms, it can help make smarter and more informed choices.

Investor dashboards could really use this technology to show predicted price trends next to the current market info. This way, both regular and big investors could get a clearer picture of where things might go in the future, helping them make smarter investment choices.

The model makes it simpler to see how expert predictions match up with what's actually happening in the real world since it provides daily updates almost right away. At the beginning of every trading day, the system works on its own and lets traders know which direction the market might head.

B. Performance in Live Scenarios

When we tested the model with real data, it made pretty good predictions no matter what was happening in the market. It wasn't always spot-on, especially during big news events that caused a lot of chaos, but it stayed really close to the actual prices. This shows that the way it learns over time in those LSTM layers works well.

C. Scalability and Automation

This system is built to grow with your needs. You can add more to it:

through different stock tickers to support multiple stocks or whole portfolios.

Connect your project to backend services like Flask or FastAPI to create APIs for forecasting.

up daily updates with tools like CRON jobs or cloud services such as AWS Lambda or Google Cloud Functions.

This system can take in real-time data, make smart predictions, and automatically handle tasks. That makes it a key part of creating a full AI-powered trading helper.

VII. Discussion

Our model's results clearly demonstrate that LSTM networks work way better than traditional methods like Random Forest or Support Vector Machines for predicting stock prices. What makes LSTM great is its ability to keep track of long-term patterns in data, which is something other models often find tough to do. Stock prices are affected by both quick changes and long-term trends, and LSTM handles both really well.

Our tests found that the model works best when we get a few key things right. For example, the length of the input window matters—a lot. We used 60 days, but if it's too short, the model doesn't see enough context. If it's too long, it might start picking up random stuff instead of useful patterns. The same goes for how deep the network is and how many LSTM units we use. These affect whether the model can spot patterns without getting too specific to the training data. Adding dropout layers really helped the model perform better on new data, and early stopping kept us from wasting time on extra training that didn't help.

A. Room for Improvement

Our model already works pretty well, but we can still improve it. We could add more tools like Bollinger Bands or MACD to help it catch smaller signals in the market. Also, using extra information such as economic reports, changes in interest rates, or even how people feel about the news might give the model a clearer picture of what's affecting the market beyond just past prices.

A good next step could be trying out hybrid models. For example, mixing LSTMs with attention layers, convolutional layers, or even transformers might help the model pay attention to the important stuff and filter out unnecessary details. Instead of just predicting one price, it could be useful to predict a range with upper and lower limits. This way, users get a clearer idea of how uncertain the prediction is, which is really important when making financial choices.

B. Real-World Considerations

Even though this model works great with test data and real-time info, we should keep in mind that no model can tell us exactly what the market will do. Things like financial crashes, political problems, or even pandemics can completely change how stocks behave in a day. Still, having a good, finely tuned model is a big help—especially if you use it along with other tools to make decisions.

The project shows that deep learning models such as LSTMs can really help with financial predictions. If you put effort

into designing them well and keep improving them, they can become strong tools for both regular investors and big trading companies.

VIII. Challenges

A big challenge in this project was dealing with how unpredictable and volatile stock market data can be. Financial data often has errors, missing info, and sudden changes caused by things like economic reports, political shifts, or worldwide crises. All these factors make it really tough to create models and make predictions, even when you're using advanced tools like LSTM.

The model has a big drawback: it only uses old price data. Sure, looking at what happened before can show some patterns, but it doesn't take into account what's happening right now, like news stories, how people feel about the market, or big economic changes that can really affect how the market moves.

Future efforts will concentrate on a few important areas to tackle these problems.

IX. Conclusion

For this project, we built and tested an LSTM model to predict stock prices. It turned out to be really good at finding patterns in past data and guessing future prices accurately. What makes this system special is that it doesn't just rely on old data; it can also adjust based on new information in real time. That makes it useful for everyday trading or analyzing finances. To make the model better, one idea is to add sentiment analysis. This would help the model understand how people feel about the market. By looking at words from news stories, financial reports, and social media, the model could figure out what the public thinks and react faster when the market changes quickly.A really interesting approach is to look at big-picture economic stuff like inflation rates, interest rate decisions, how much the economy is growing (GDP), and employment reports. All these bigger economic signs give important background info that can help the model make smarter guesses.Looking ahead, trying out more advanced designs is part of our plan. Methods that bring together several models, like ensemble learning, might help us get better results and make our system stronger. Also, models based on attention, like Transformers, could perform even better because they can focus on the most important parts of the information they're processing.In the end, adding support for forecasting across different assets and timeframes would make the model more useful. It would let the system predict trends for various stocks, industries, or indexes over different periods like days, weeks, or months. This would make the whole thing more adaptable and valuable in real-life situations.We're working on solving these issues so we can build a better forecasting system. One that works well no matter what's happening in the market and gives us the info we need to make good decisions. The way this real-time prediction system works proves that these models aren't just for academics—they can actually be used in real-world finance. It's fast, so it gets the newest data and gives fresh predictions quickly, which is really helpful for investors and analysts.Of course, we're just getting started. There's still a lot we can do to make the model stronger and more flexible. For future work, we could try adding more tools that help analyze the market, test it on different stocks or industries, and see if it can predict what happens over several days. Also, including things like economic news, interest rates, or how people feel about the market could help the model understand the bigger picture better. This study suggests that deep learning, especially LSTM networks, could be really useful for predicting finance if it's carefully designed and used in real-time.

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